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HUMAN-COMPUTER INTERACTION
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Chapter 2

The Human

The Human

- In order to design something for someone, we need to understand their capabilities and limitations.
  - Help us to know what people find easy and how we can help them by encouraging these things
- To understand cognitive psychology which have a bearing on the use of computer systems:
  - how humans ***perceive*** the world around them, how they **store** and **process** information and solve problems, and how they **physically manipulate** objects
- To think of a simplified *model* of what is actually going on

- Use the analogy of the user as an information processing system, and make the analogy closer to that of a conventional computer system.
  - Information comes in, is stored and processed, and information is passed out.
- Three core components of the system : input–output, memory and processing.
- In the human, we are dealing with an intelligent information-processing system, and processing therefore includes problem solving, learning, and, consequently, making mistakes

NPUT–OUTPUT CHANNELS

- A person’s interaction with the outside world occurs through information ***being received and sent***: input and output.
- In an interaction with a computer the user receives information that is output by the computer, and responds by providing input to the computer
- There are five major senses: ***sight, hearing, touch, taste and smell***
- vision, hearing and touch are central to HCI

NPUT–OUTPUT CHANNELS

“Imagine using a personal computer (PC) with a mouse and a keyboard. The application you are using has a graphical interface, with menus, icons and windows. In your interaction with this system you receive information primarily by sight, from what appears on the screen. However, you may also receive information by ear: for example, the computer may ‘beep’ at you if you make a mistake or to draw attention to something, or there may be a voice commentary in a multimedia presentation. Touch plays a part too in that you will feel the keys moving (also hearing the ‘click’). You yourself send information to the computer using your hands, either by hitting keys or moving the mouse “.

the human

- Information i/o …
  - visual, auditory, haptic, movement
- Information stored in memory
  - sensory, short-term, long-term
- Information processed and applied
  - reasoning, problem solving, skill, error
- Emotion influences human capabilities
- Each person is different

Vision

Two stages in vision

- physical reception of stimulus
- processing and interpretation of stimulus

The Eye - physical reception

- mechanism for receiving light and transforming it into electrical energy
- light reflects from objects
- images are focused upside-down on retina
- retina contains rods for low light vision and cones for colour vision
- ganglion cells (brain!) detect pattern and movement

Interpreting the signal (cont)

- Brightness
  - subjective reaction to levels of light
  - affected by luminance of object
  - measured by just noticeable difference
  - visual acuity(clarity) increases with luminance
- Colour
  - made up of hue, intensity, saturation
  - 8% males and 1% females colour blind

Reading

- Several stages:
  - visual pattern perceived
  - decoded using internal representation of language
  - interpreted using knowledge of syntax, semantics, pragmatics

- Reading involves saccades and fixations
- Perception occurs during fixations
- Word shape is important to recognition
- Negative contrast improves reading from computer screen

Hearing

- Provides information about environment:
  distances, directions, objects etc.
- Physical apparatus:
  - outer ear – protects inner and amplifies sound
  - middle ear – transmits sound waves as vibrations to inner ear
  - inner ear – chemical transmitters are released and cause impulses in auditory nerve
- Sound
  - pitch – sound frequency…how high or low tone
  - loudness – amplitude(how soft or loud)
  - timbre – type or quality

Hearing (cont)

- Humans can hear frequencies from 20Hz to 15kHz
  - less accurate distinguishing high frequencies than low.
- Auditory system filters sounds
  - can attend to sounds over background noise.
  - for example, the cocktail party phenomenon.

Touch

- Provides important feedback about environment.
- May be key sense for someone who is visually impaired.
- Stimulus received via receptors in the skin:
  - thermoreceptors – heat and cold
  - nociceptors – pain
  - mechanoreceptors – pressure (some instant, some continuous)
- Some areas more sensitive than others e.g. fingers.
- ***Kinethesis*** - awareness of body position
  - affects comfort and performance.

Movement

- A simple action such as hitting a button in response to a question involves a number of processing stages.
- The stimulus (of the question) is received through the sensory receptors and transmitted to the brain.
- The question is processed and a valid response generated. The brain then tells the appropriate muscles to respond.
- Each of these stages takes time, which can be roughly divided into reaction time and movement time.
- Time taken to respond to stimulus:
  reaction time + movement time

Movement

- Speed and accuracy of movement are important considerations in the design of interactive systems, primarily **in terms of the time taken to move to a particular target on a screen.**
- The target may be a button, a menu item or an icon, for example.
  - The time taken to hit a target is a function of the **size** of the target and the **distance** that has to be moved.
  - Since users will find it more difficult to manipulate small objects, **targets should generally be as large as possible and the distance to be moved as small as possible**

Memory

Memory game

- "I went to the market and bought a lemon, some oranges, bacon . . .' or 'I went to the zoo and saw monkeys, and lions, and tigers . . .' and so on.
- As the list grows objects are missed out or recalled in the wrong order and so people are eliminated from the game.

- Three types of memory or memory function: ***sensory buffers, short-term memory or working memory, and long-term memory***

Memory

There are three types of memory function:

Sensory memories



Attention

Short-term memory or working memory



Rehearsal

Long-term memory

Selection of stimuli governed by level of arousal.

sensory memory

- Information is passed from sensory memory into short-term memory by attention, thereby **filtering the stimuli to only those which are of interest at a given time**.
- Attention is the **concentration of the mind** on one out of a number of competing stimuli or thoughts
- Cocktail party phenomenon mentioned earlier: we can attend to one conversation over the background noise, but we may choose to switch our attention to a conversation across the room if we hear our name mentioned.

- Information received by sensory memories is quickly

sensory memory

- Buffers for stimuli received through senses
  - iconic memory: visual stimuli
  - echoic memory: aural stimuli
  - haptic memory: tactile stimuli
- Examples
  - moving a finger in front of the eye. Can you see it in more than one place at once?
  - echoic memory is evidenced by our ability to ascertain the direction from which a sound originates
- Continuously overwritten

Short-term memory (STM)

- Short-term memory or working memory acts as a ‘scratch-pad’ for temporary recall of information
- *Ex: Consider reading*: In order to comprehend a sentence, you need to hold in your mind the beginning of the sentence as you read the rest.
  - rapid access ~ 70ms
  - rapid decay ~ 200ms
  - limited capacity - 7± 2 chunks

Short term memory….

- it also decays rapidly, meaning that information can only be held there temporarily, in the order of 200 ms.
- Short-term memory also has a limited capacity
- The average person can remember 7 ± 2 digits
- Look at the following number sequence:

212348278493202

- Now write down as much of the sequence as you can remember.
- Did you get it all right? If not, how many digits could you remember?

Short term memory…..

- Now try the following sequence:

  44 113 245 8920

- Here the digits are grouped or *chunked*.
- A generalization of the 7 ± 2 rule is that we can remember 7 ± 2 *chunks* of information.
- Therefore chunking information can increase the short-term memory capacity

Short term memory….

- The successful formation of a chunk is known as *closure*.
- This process can be generalized to account for **the desire to complete or close tasks held in short-term memory.**
- If a subject fails to do this or is prevented from doing so by interference, the subject is liable to lose track of what she is doing and make consequent errors.

Short term memory

Design Focus of Closure

"Closure gives you a nice 'done it' when we complete some part of a task. At this point our minds have a tendency to flush short-term memory in order to get on with the next job. Early automatic teller machines (ATMs) gave the customer money before returning their bank card. On receiving the money the customer would reach closure and hence often forget to take the card. Modern ATMs return the card first! "

Short Term Memory…HCI Design rationale

Design Rationale:

- **Allow for Reversibility (Undo):** Users will forget the exact state they were in.
- A robust "Undo" function is a safety net for lapses in STM.
- **Keep Important Information in View:** If a user is filling out a multi-step form, keep key context or the form's overall structure visible so they don't have to remember what step they are on.
- **Provide Consistent Feedback:** If an action takes more than a second or two, provide a progress indicator. This reassures the user their request is being processed, so they don't have

Short term memory….

Design Rationale:

- STM is perhaps the most critical constraint for interface design.
- We must avoid overloading the user's STM.
- **Chunking:** Present information in meaningful groups. A phone number is written as 123-456-7890 (three chunks) not 1234567890 (ten chunks).
- **Simplify Interfaces:** Minimize clutter. Every extra button, icon, or piece of text is competing for one of those precious 7 slots.

Long Term Memory

- If short-term memory is our working memory or ‘scratch-pad’, long-term memory is our main resource.
- Here we store factual information, experiential knowledge, procedural rules of behavior – in fact, everything that we ‘know
- Information from short-term memory is stored in long-term memory by r**ehearsal**.
- The repeated exposure to a stimulus or the rehearsal of a piece of information transfers it into long-term memory.

Long-term memory (LTM)

- Repository for all our knowledge
  - slow access ~ 1/10 second
  - slow decay, if any
  - huge or unlimited capacity

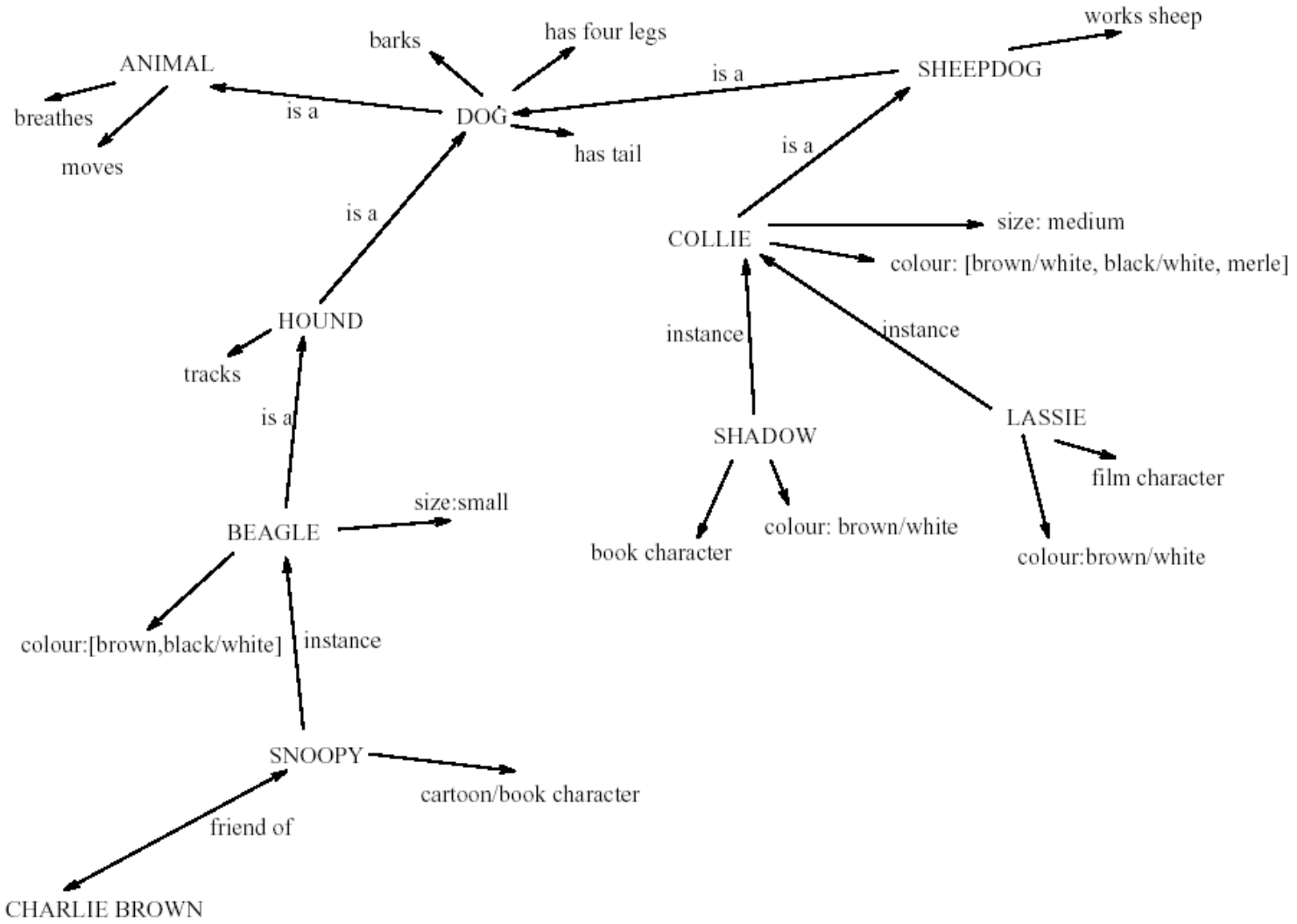
- Two types
  - episodic – stores serial memory of events
  - semantic – structured memory of facts, concepts, skills

-semantic LTM derived from episodic LTM such that we can learn new facts or concepts from our experiences.

Long-term memory (cont.)

- Semantic memory structure
  - provides access to information
  - represents relationships between bits of information
  - supports inference
- Model: semantic network
  - inheritance – child nodes inherit properties of parent nodes
  - relationships between bits of information explicit
  - supports inference through inheritance

LTM - semantic network



Models of LTM - Frames

- Information organized in data structures
- Slots in structure instantiated with values for instance of data
- Type–subtype relationships

DOG

Fixed
legs: 4

Default
diet: carniverous
sound: bark

Variable
size:
colour

Models of LTM - Production rules

Representation of procedural knowledge.

Condition/action rules

if condition is matched

then use rule to determine action.

IF dog is wagging tail
THEN pat dog

IF dog is growling
THEN run away

LTM - Storage of information

- rehearsal
  - information moves from STM to LTM
- total time hypothesis
  - amount of info retained proportional to rehearsal time
- distribution of practice effect
  - Recall is optimized by spreading learning over time
- structure, meaning and familiarity
  - Information is easier to remember if grouped

LTM - Forgetting

decay
- information is lost gradually but very slowly

interference
- new information replaces old: retroactive interference
- Eg: if you change telephone numbers, learning your new number makes it more difficult to remember your old number

proactive inhibition
- old may interfere with new:
- Eg: when you find yourself going to your old house rather than your new one.

Affected by emotion
- can subconsciously `choose' to forget eg positive over negative

LTM - retrieval

recall

- In recall the information is reproduced from memory
- recall can be assisted by the provision of retrieval cues, which enable the subject quickly to access the information in memory. One such cue is the use of categories. e.g. categories, imagery

recognition

- In recognition, the presentation of the information provides the knowledge that the information has been seen before.
- Recognition is **the less complex cognitive**

LTM – HCI Design rationale

- **Recall:** Retrieving information from LTM with little to no external cue (e.g., "What was the name of that restaurant we went to last year?"). This is *hard*.

- **Recognition:** Identifying information from a set of presented options (e.g., seeing a list of restaurants and spotting the one you went to). This is *easy*.

  - **Design Rationale:** Our primary goal as designers is to favor **recognition over recall**.

LTM – HCI Design rationale

- **Use Menus and Icons:** Instead of asking users to recall a command name (e.g., "Format Painter"), present them with a toolbar of icons they can recognize.
- **Provide Suggestions & Autocomplete:** Search bars are the perfect example. As you type "New..." Google suggests "New York," "New Orleans"—you just have to recognize the correct one.
- **Maintain Consistency:** Use consistent icons, labels, and layouts across your application and, where possible, follow platform conventions. That is stored in users' LTM. Changing it to a different, unfamiliar icon forces problematic

LTM – HCI Design rationale

- **Interference (The "Cluttering" of Memory):**
  - Interference occurs when memories compete and get mixed up
- **Design Rationale:** We must design interfaces that minimize memory competition, especially during learning and transition periods.
- **Leverage Existing Mental Models:** When introducing a new feature, relate it to something the user already knows.
  - For example, calling a new cloud-syncing service

LTM – HCI Design rationale

- **Decay and Strengthening through :**
  - While LTM is more permanent, memories can decay or become harder to access if not used. The strength of a memory is reinforced by repetition and meaningful association.
- **Design Rationale:**
  - We should design for both novice users (whose memories for our interface are weak or non-existent) and expert users (who have strong, reinforced memories).

LTM – HCI Design rationale

- **Provide Keyboard Shortcuts for Experts:**
  - Experts who use a function frequently strengthen that memory. Keyboard shortcuts allow them to bypass recognition-based menus and use their fast, recalled knowledge.
- **Use Intuitive Metaphors:**
  - Associating a "Trash Can" with deletion creates a strong, meaningful link in LTM, making the concept easier to learn and remember.
- **Maintain Stability:**
  - Avoid frequent, drastic redesigns. A major overhaul can make a user's well-practiced, strong memories for the old interface suddenly obsolete, forcing them back into a novice state.

Errors and mental models

Types of error

- slips
  - right intention, but failed to do it right
  - causes: poor physical skill,inattention etc.
  - change to aspect of skilled behaviour can cause slip
- mistakes
  - wrong intention
  - cause: incorrect understanding

    humans create mental models to explain behaviour.
    if wrong (different from actual system) errors can occur

Emotion

- Various theories of how emotion works
  - James-Lange: emotion is our interpretation of a physiological response to a stimuli
  - Cannon: emotion is a psychological response to a stimuli
  - Schacter-Singer: emotion is the result of our evaluation of our physiological responses, in the light of the whole situation we are in
- Emotion clearly involves both cognitive and physical responses to stimuli

Emotion (cont.)

- The biological response to physical stimuli is called *affect*

- Affect influences how we respond to situations
  - positive ⎯ creative problem solving
  - negative ⎯ narrow thinking

"Negative affect can make it harder to do even easy tasks; positive affect can make it easier to do difficult tasks"

(Donald Norman)

Emotion (cont.)

- Implications for interface design
  - stress will increase the difficulty of problem solving
  - relaxed users will be more forgiving of shortcomings in design
  - aesthetically pleasing and rewarding interfaces will increase positive affect

Individual differences

- long term
  - sex, physical and intellectual abilities
- short term
  - effect of stress or fatigue
- changing
  - age

Ask yourself:
will design decision exclude section of user population?